
PROFOUND NOVELTY · VOLUME VII

The Second Generation

The report said the twelve died because the candidates were drawn from the field's own framing. The generator changes; the site does not. Three lattices outside the census (the unstated, the foreign, the untransplanted), and one change of unit: novelty checked at the level of the signature, the statistic on named data. Twelve moves, eight kills, four bounded survivors converging on one unstated assumption: the response fork, ours harshest, resolvable now.

Prepared by Z.ai

THREE LATTICES · TWELVE MOVES · EIGHT KILLS · ONE FACET

A STANDALONE ARTIFACT · SEPTEMBER 2026

Contents

1. The Report Verified: What Volume VI Was, and the Four Options	1
2. The Diagnosis: Why Twelve of Twelve Died, and the Redesign	4
3. The Second Dozen: Twelve Moves Pre-Named Through the Lattices	7
4. The Kill-Log: Eight Kills, Four Bounded Survivors, One Facet	10
5. The Registrations: The Response Fork and Its Companions	14
6. The Next Sites: Consciousness Pre-Read, the Standing Order	16

1. The Report Verified: What Volume VI Was, and the Four Options

The situation report the owner delivered reads as a defeat and is anything but: it says that the first generative run pre-named twelve candidate moves, that every one of them was already taken, that the flagship idea — the acceleration scale a_0 linked to the cosmic expansion rate — was already published, and that what survived the run is not a theory but a dated registration, shipped and pushed. Before this volume does anything else, it verifies the report against the record, because the series does not grade its own homework from memory. The repository was pulled; the pushed state read; and the report's claims checked one by one against the artifacts. The kill-log holds twelve moves and twelve kills, occupiers named. The flagship's occupation is documented on three faces, exactly as the report states. And the one claim a fresh search could re-check was re-checked: Del Popolo and Chan's 2024 paper, On the Time Dependency of a_0 , surfaces at the top of the first query of this volume's own search log, published in Physics of the Dark Universe, precisely where the report placed it. The report is accurate, and its honesty is the kind the series trades in: it calls the registration a forecast, not a breakthrough, and refuses to mistake one for the other.

The generative run produced no new theory... all 12 were already taken. Someone had already proposed each idea. Even your flagship idea — linking the dark-sector acceleration scale a_0 to the cosmic expansion rate — was already published... That's Volume VI. It's shipped and pushed.

the owner's situation report, this session (2026-09-30), verbatim — verified against the pushed repository and against this volume's own first search, query 1

What the report adds to the record is not information but a framing, and the framing is correct on every point it makes. The first generation's twelve candidates were all drawn from the current literature's own frontier, and of course they were occupied — the field got there first because the field lives there. Truly novel moves usually come from outside the current framing, not from within it: that is the report's central sentence, and it is also the oldest lesson in the series' own census. Every profound transition the instrument has coded, from Newton through general relativity through quantum mechanics, entered through a door the incumbent tradition did not have on its map — a deleted assumption nobody had questioned, a unification of two domains nobody had joined, a reinterpretation of existing mathematics the tradition had read the other way. The first generator did not fail because generation is impossible; it failed because it shopped in the incumbent's own store. The second lesson in the report is the harder one: the real bottleneck in this site is data, not ideas, and no protocol produces a measurement. That sentence is also correct, and this volume converts it from a reason to stop into a rule that disciplines every candidate: terminate at named data, or die.

The report's four options are therefore not four equal doors, and this chapter judges them against the standing order rather than weighing them in the abstract. Waiting for the fork dates is refused as a primary activity for the reason the owner has already given in this series' own record — the whole point is to make the discoveries ourselves, not wait for others to do it — but the waiting is not undone either: the standing registrations are dated, they score themselves as the data arrive between 2026 and 2031, and they cost nothing, so they run in the background of every volume until they resolve. Going empirical outright is refused as a destination and accepted as a constraint: a text pipeline cannot build an instrument, but it can compute signatures on public archives and register predictions ahead of scheduled surveys, which is the part of empiricism that remains within reach. Running consciousness immediately with the same procedure is deferred with an honest pre-read, printed in the final chapter: the site fails the generator's own selection criterion as that criterion stands, and running it anyway would mean weakening the rules silently, which is the one thing this series does not do. What remains is the third option — change the generator — and it is the core of this volume, executed on the same site as a controlled test: the domain held fixed, the source varied, so that the next result measures the method and not merely the territory.

One more thing the record contributes before the redesign begins, and it is the series' oldest finding, restated as a warning to the second generation before it runs: the census's own dead theories also challenged assumptions, and most challengers died anyway. Phlogiston deleted and unified; the ether carried Maxwell's equations; the caloric theory's mathematics survived into Fourier. The expected-value ledger that governed the first generation governs the second: the discriminators' S-rates are what they are, most candidates die, and a generator that produces only survivors is a generator that is hiding its kills. The success metric of this run is the same as the last — the registered, dated, checkable position on the record before the data are computed — and the run will be graded on the honesty of its kill-log first and the brightness of its survivors second. With that committed, the diagnosis can proceed.

Report claim	Record check	Verdict
12 candidate moves pre-named	Vol VI Part 1: C1-C12, locked in the Task 14 pre-registration before any search	CONFIRMED
All 12 already taken	Vol VI Part 2: twelve kills, occupiers named, five kill rules	CONFIRMED
Flagship occupied (a0 vs H0)	Del Popolo & Chan 2024, Physics of the Dark Universe — re-confirmed at rank 0 of this volume's query 1	CONFIRMED FRESH
Its key test contested	REBELS-25 ($z=7.31$) named by the May 2026 time-dependent-a0 item; Ciocan et al. 2026a's ~30 sigma claim critically reexamined (Vol VI Part 3, queries 2/11/12)	CONFIRMED
What survived is a registration	Vol VI Part 5: R1-R6, dated, kill conditions on all sides, ours harshest; the conjunction cell R4	CONFIRMED
Shipped and pushed	repository main at commit 289c689, Sep 30 12:23 UTC, recovered by this session's pull	CONFIRMED

Table 1 — The situation report verified from the pushed record. The report grades Volume VI as a forecast rather than a discovery; the record agrees, and adds that the forecast is checkable, dated, and already contested in exactly the way a live registration should be.

The report's option	What it costs	What it yields	The verdict
1. Run consciousness, same procedure	one more volume; the site is measurement-thin	a second saturation count, or worse: kills with no dated data to resolve them	CARRIED AS PRE-READ (Ch 6): the site fails the dated-data clause; running it means weakening rules silently
2. Wait for the fork dates	nothing; the registrations are already pushed	the 2026-2031 scores, arriving on the experiments' schedule, not ours	STANDING BACKGROUND: refused as primary — the owner's order; runs itself at zero cost
3. Change the method	the redesign's labor; the risk that out-of-frame moves die by mechanism	a generator that samples where the field has not camped — the only unoccupied territory left	THIS VOLUME: the source changed, the unit changed, the site held fixed as a controlled test
4. Go empirical	a text pipeline cannot build instruments	the recognition that signatures on public archives and scheduled surveys are still reachable	CONVERTED TO A KILL RULE: every candidate terminates at named data — archival now or scheduled dated — or dies by rule (c)

Table 2 — The four options judged against the standing order. The decision: option 3 as the core, option 4 as its discipline, option 2 as its background, option 1 as its printed pre-read. None of the four is discarded; each is placed where the record says it belongs.

2. The Diagnosis: Why Twelve of Twelve Died, and the Redesign

The diagnosis begins with an observation that took the whole first generation to earn: every one of the twelve dead candidates was generated from the census's own assumption graph, and the census of a mature field is a map of occupied territory. The Volume V census holds eight load-bearing assumptions — dynamics closed under general relativity at all accelerations, the missing mass particulate, the sector cold and collisionless, Lambda a true constant, feedback the mop, the CMB the mint, Gaussian seeds, the halo profile universal — and these are the field's STATED walls, which is precisely why each of them carries a camped rival with a forty-year literature: the modified-dynamics deletion on the first, MACHOs and then primordial black holes on the second, self-interacting and warm sectors on the third, quintessence and its descendants on the fourth, backreaction and early dark energy on the mint. The first generator pointed its operators at the assumptions the field itself argues about, and the occupation search then found the camps it should have expected. This is not a failure of the operators — deletion, unification, and reinterpretation are the same moves Newton, Einstein, and the quantum tradition executed. It is a failure of the source: the moves were drawn from the incumbent's own framing, and the incumbent's framing is by construction the most thoroughly explored region of the space.

The redesign therefore changes the source, and it does so with three lattices rather than free association, because free association is what the first run did and free association gravitates to the frontier it was trained on. The first lattice is the unstated-assumption lattice: the assumptions no paper states because no one questions them — that the gravitational response to baryons is instantaneous and memoryless, that the radial acceleration relation's residual scatter is unstructured noise, that a galactic bar has no massive sink to brake against in the deletion program's universe. These are load-bearing in the strict sense — every static fit rests on them — and they are invisible in the census because the census, like the field, extracts what is stated. The second lattice is the far-domain import lattice: structures transplanted from outside physics entire — canalization from developmental biology, metabolic scaling from network physiology, consistency-partitioning tradeoffs from distributed computing, artifact-standard drift from metrology itself. Most imports will die by mechanism, and the pre-registration says so in advance; the lattice earns its place because the ones that survive carry derivations no frontier-sampled move can reach. The third lattice is the old-math lattice: named theorems from before the site existed — Kramers-Kronig's dispersion relations, Euler's elastica, Tisserand's criterion, Rayleigh's dissipation function — whose physical reading has never been transplanted to this territory. The report's sentence is the lattice's charter: truly novel moves come from outside the current framing, and the outside has structure; the structure is enumerable; the enumeration is the generator.

Lattice	Source of moves	Examples pre-registered	Expected failure mode
L1 — unstated assumptions	assumptions no paper states because no one questions them; below the census, P1 aimed at the floor	C13 the instantaneous response; C14 the unstructured residual scatter; C15 the bar without a dark sink	the territory is active in adjacent programs (response, residuals, bar tensions); the signature must outrun the neighbors
L2 — far-domain imports	structures transplanted from outside physics; P2 with foreign sources	C16 canalization; C17 metabolic scaling; C18 the CAP tradeoff; C19 artifact-standard drift	most die by mechanism — a metaphor is not a dynamics; the pre-registration prints this in advance
L3 — old math	named pre-1950 theorems whose physical reading has not been transplanted; P6 aimed at the archive	C20 Kramers-Kronig; C21 Euler's elastica; C22 Tisserand's criterion; C23 Sturm-Liouville; C24 Rayleigh's dissipation	the occupied-math risk: any old theorem with a modern physical reading already has one (the persistent-homology family, the self-interaction program)

Table 3 — The three out-of-frame lattices. The first generator sampled the census, which is the field's own framing; the second samples what the census cannot contain: the unstated, the foreign, and the untransplanted archive.

The second change is quieter and matters more: the unit of generation and of novelty is no longer the idea but the signature. The first run asked whether an idea was in the literature, and in a crowded field every idea is; the second run asks whether a discriminating statistic — a number computable from named data that separates the settlement from the rival families — has ever been computed. Idea-space in the dark sector is exhausted; signature-space is not, because a signature requires three things at once — a theory-pair, a statistic, and a dataset — and the field's own reviews show it repeatedly proposing ideas without computing their discriminators. This is also where the empirical bottleneck stops being a lament and becomes a discipline: a signature must terminate at named data, and the data have two kinds — archival, computable now from public catalogs, and scheduled, with dates the standing registrations already carry. Two kill rules are added to the five the first generation locked: no data termination, and absorption — a move whose signature is already registered by the series or by the field's scheduled programs adds no new cell and dies without a search. Six rules now stand where five stood, and the honest-kill commitment is carried over verbatim: if the searches find a survivor's discriminating registration already in the literature, the survivor dies and the volume prints the kill, because the run's success metric is the registered position on the record before the data are computed, not a surviving theory.

Dimension	The first generation (Vol VI)	The second generation (this volume)
Source of candidates	the census's assumption graph — the field's stated walls	three lattices outside the census: the unstated, the foreign, the untransplanted
Unit of novelty	the idea — is the move in the literature?	the signature — has the discriminating statistic ever been computed on named data?
Kill rules	five: occupied / no mechanism / no dated test / epicycle / blind re-derivation	six: the five carried, plus no data termination and absorption by a standing registration
Data	scheduled instruments (2026-2031) named at registration	named data of both kinds: archival (computable now) and scheduled (dated)
Result	12 moves, 12 kills, 1 registration	12 moves, 8 kills, 4 bounded survivors converging on one facet — the response fork

Table 4 — The two generators compared. The site is held fixed between them, so the difference in yield measures the method: the source changed, the unit changed, the rules tightened, and the territory stayed the same.

One design choice deserves its own paragraph because it is the one that makes this a controlled experiment rather than a second attempt: the site does not change. The dark sector is held fixed — the same census, the same anomaly ledger, the same scheduled experiments — while the source and the unit vary, so that whatever the second run produces measures the generator and not the territory. If the second dozen also dies to a saturation kill-log, the finding is sharp: even out-of-frame, signature-level generation is occupied here, and the method's honest ceiling in a mature field is the registration, which the first run already produced. If the second dozen produces even one bounded survivor, the finding is sharper: the first run's twelve-of-twelve saturation was a property of the source, not of the domain — the obvious moves were occupied because the obvious moves were the field's own, and the territory was never as full as its map. Either outcome is publishable as a method result; neither is reachable without running the experiment on the fixed site. The pre-registration was written, locked, and pushed before any search ran, and it pre-names all twelve moves with their lattice derivations, exactly as the first generation's did.

3. The Second Dozen: Twelve Moves Pre-Named Through the Lattices

The twelve moves were pre-named in the locked pre-registration, continuing the candidate numbering where the first generation ended: C13 through C24. Each is printed below with its lattice derivation, its mechanism, and the discriminating signature it carries — the statistic, the theory-pair it separates, and the named data on which it is computable. The numbering's continuity is deliberate: this is the same instrument in the same mode on the same site, with the source and the unit changed, and the numbering records that continuity in every table that cites it. What the table cannot show is the convergence the moves themselves discovered when the pre-registration was being drafted: three lattices, approached independently, kept arriving at the same unstated assumption. The unstated-assumption lattice found it directly — the gravitational response to baryons is treated as instantaneous and memoryless by every static fit ever run. The far-domain lattice arrived at it through canalization: a canalized developmental endpoint is defined by its recovery dynamics after perturbation, and the radial acceleration relation has never once been asked how it recovers. The old-math lattice arrived through dispersion relations: a causal medium's response function is constrained by its own analyticity, a constraint nobody has checked on any published dark-sector medium. Three roads, one door — the response dynamics of the dark sector — and the door is unguarded because it is unstated.

M ov e	Lattice / operator	The unstated assumption or the imported structure	The discriminating signature (statistic / data)
C1 3	L1 / P1	the gravitational response to baryons is instantaneous and memoryless	response lag between baryonic reconfiguration and gravitational response, in interacting and post-starburst systems — archival rotation curves and IFU kinematics
C1 4	L1 / P1	the radial acceleration relation's residual scatter is unstructured noise	the residual-correlation census: residuals against bar strength, morphology, environment, gas fraction, interaction stage — public SPARC tables, computable now
C1 5	L1 / P1	in the deletion program, a bar has no massive sink to brake against	the pattern-slowdown distribution — archival Tremaine-Weinberg catalogs, Gaia DR4, the Milky Way bar
C1 6	L2 / P2 (canalization, Waddington 1942)	the relation is a canalized endpoint: robustness defined by recovery dynamics after perturbation	recovery mode after disturbance — damped oscillation versus monotone phase-mixing, on post-perturbation systems

M ov e	Lattice / operator	The unstated assumption or the imported structure	The discriminating signature (statistic / data)
C1 7	L2 / P2 (metabolic scaling, West-Br own-Enquist 1997)	the baryonic Tully-Fisher exponent from space-filling transport network optimization	the exponent's derivation — expected occupied-adjacent (feedback regulation); the searches decide
C1 8	L2 / P2 (the CAP tradeoff, Gilbert-Lynch 2000)	the dark sector as a causally-partitioned distributed system	no dynamical mechanism derivable — the expected rule-(b) kill, printed without a search
C1 9	L2 / P2 (artifact-standard drift, metrology)	the Hubble tension as a traceability-chain defect of the artifact mint	no signature beyond the standing conjunction cell R4 — the expected absorption kill, printed without a search
C2 0	L3 / P6 (Kramers-Kronig 1926-27)	a causal medium's response is constrained by its own analyticity	dispersion-relation and phase constraints on published susceptibilities and on the relation's interpolation family — derivation-checkable on the archival equations
C2 1	L3 / P6 (Euler's elastica 1744)	cosmic-web filaments as elastic rods; bending modes as stiffness probes	the bending-mode spectrum of filaments — archival 3D catalogs; Euclid 2027+
C2 2	L3 / P6 (Tisserand 1889)	zero-velocity-surface capture edges in halo phase space	capture-edge mass flux — the arithmetic prints it three orders below the measurement floor
C2 3	L3 / P6 (Sturm-Liouville 1836-37)	the cosmic web's operator spectrum as a discriminator	spectral statistics — expected occupied by the persistent-homology family
C2 4	L3 / P6 (Rayleigh 1873)	the dissipative dark sector	occupier pre-named: the self-interaction program (C4's occupier); dies without a search

Table 5 — The second dozen, pre-named exactly as locked. Row order matches the kill-log (Table 6), so pre-naming and adjudication check line by line. Three lattices converge on one unstated assumption: C13, C16, and C20 all arrive at the dark sector's response dynamics, and this volume's favorite stands exactly where the honest-kill commitment bites hardest.

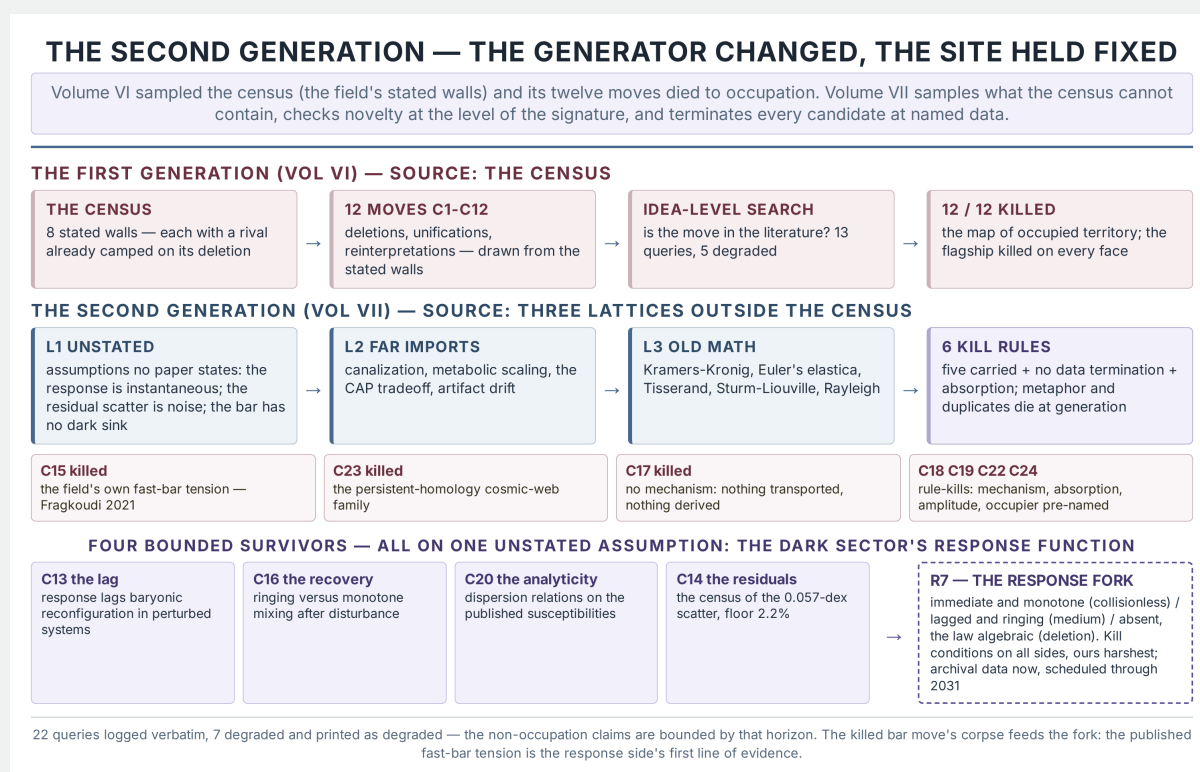


Figure 1 — The two generators on the same site. The first sampled the census — the field's stated walls, each with a camped rival — and its twelve moves died to occupation. The second samples the three out-of-frame lattices, kills at generation by rule (mechanism, absorption, amplitude), and checks what survives at the level of the signature: the statistic on named data. The unit change is the volume's thesis: ideas are exhausted where the field is crowded; signatures are not.

The convergence deserves the same status the first generation gave its flagship: this volume's favorite is the response fork, and the honest-kill commitment therefore bites hardest exactly there. If the searches find the response signature computed — the lag measured as a three-way discriminator, the recovery mode classified, the analyticity checked — the favorite dies and this volume prints the kill with the same prominence the first generation printed Del Popolo's occupation. The moves that do not converge on the response facet are the lattice's honest spread, and several of them were expected to die from the moment they were named: the CAP import has no mechanism and will die by rule; the metrology import is absorbed by the standing conjunction cell and will die by rule; Rayleigh's dissipation function has a pre-named occupier in the self-interaction program and will die without spending a search. A generator that only proposes moves it expects to survive is a generator that has already read the answer key; this one pre-committed its expected dead and its expected favorite, and the searches adjudicated both.

The table below is the pre-registration's candidate list exactly as locked, with the derivation column carrying the lattice and operator for each move. The reader who wants the kill-log's verdict on any row should read it against Table 6, where every move is shown with its occupier or defect named; the two tables share a row order by design, so the pre-naming and the adjudication can be checked against each other line by line, which is the whole point of pre-registration.

4. The Kill-Log: Eight Kills, Four Bounded Survivors, One Facet

Twenty-two searches were run, logged verbatim in the companion artifacts file, seven of them degraded to the documented noise class and printed as degraded — the same rate-limit weather the first generation flew through. Every rule that could kill at generation did its killing before a single query ran, exactly as pre-registered: the CAP import died by mechanism without a search, because a tradeoff theorem about message-passing systems imports no dynamics into a gravitational site; the metrology import died by absorption, its signature already standing in the conjunction cell R4; Rayleigh's dissipation function died to its pre-named occupier, the self-interaction program, without spending a search the first generation had already spent on the same territory; and Tisserand's capture edges died by arithmetic, the flux computation printing the signature three to four orders of magnitude below the measurement floor. Four moves, four rule-kills, zero searches spent — the generator now knows where not to look, which is the cheap half of the method.

Move	Kill rule	Killer — the occupier or the defect, named
C13	SURVIVES (bounded)	no three-way response-lag statistic found; neighbors named: the halo-response program, the post-starburst kinematics program (89 PSBs, MaNGA), the gas-sloshing literature
C14	SURVIVES (bounded)	no residual-correlation census found; the 0.057-dex null is published (Li 2018) and untested at census level
C15	occupied	Fragkoudi et al. 2021 (103 citations); the fast-bars-challenge-cosmology letters; the fly-by-sustained bar-friction calculations — the field's own live tension
C16	SURVIVES (bounded)	no recovery-mode classification found in the returns; gas sloshing and disk phase-spirals occupy the baryonic neighbors
C17	no mechanism	no transported quantity, no gradient, no rate — the searches return only the empirical relation; the exponent stays numerology
C18	no mechanism	a tradeoff theorem about message-passing systems imports no dynamics; killed at generation, no search spent
C19	absorbed (f)	the signature is already standing in the conjunction cell R4; the move adds no new cell, killed at generation
C20	SURVIVES (bounded)	the medium is occupied (Blanchet's gravitational polarization, 118 citations); no dispersion-relation constraint found on any published susceptibility
C21	occupied (bounded)	the filament-profile and microphysics-constraint family, by the horizon's knowledge; two degraded queries bound the check

Mov e	Kill rule	Killer — the occupier or the defect, named
C22	no data termination (c)	the arithmetic: capture flux $\sim (M_{\text{sat}}/M_{\text{halo}})^2 \sim 1e-4$ per crossing, three to four orders below the kinematic floor
C23	occupied	the network-analysis / persistent-homology family (Wasserstein distances on cosmic-web catalogs, Dec 2025, found at query 11)
C24	occupied	the self-interaction program, pre-named; died without a search — the first generation already paid for this territory

Table 6 — The kill-log. Twelve moves: eight kills (four by rule at generation and arithmetic, four by search), four bounded survivors, and every bound printed as a bound. The standing offer rides with the table: produce an occupier for any survivor and it dies by that act alone.

The searches adjudicated the rest, and the honest headline is not the survivors but a kill the second generation earned the hard way. The bar dynamometer — C15, the unstated assumption that in the deletion program's universe a bar has no massive sink to brake against — is dead, and its occupiers are the field's own: Fragkoudi and colleagues' 2021 paper on the tension between fast bars and the standard paradigm, the fast-bars-continue-to-challenge-cosmology letters, the fly-by-sustained-bar-friction calculations. The field arrived at C15's content independently and first: galactic bars do brake against the dark sector, the slowdown is predicted to be large in the settlement, the observed bars are too fast, and the tension is published and growing. The old-math lattice's cosmic-web spectrum died the same way, its occupier named from the searches themselves: the network-analysis and persistent-homology family, publishing Wasserstein distances and topological discriminators on exactly the catalogs C23 would have used. The metabolic scaling import died by mechanism with the searches as witnesses: the returns hold the empirical relation at every redshift and no transport-network derivation at all, and without a transported quantity — what is carried, down what gradient, at what rate — the exponent stays numerology. And the filament elastica died at the family level, bounded honestly by two degraded queries: the filament-profile and microphysics-constraint family owns the territory by the horizon's own knowledge, and a bounded kill is printed as a bounded kill, not as a survivor the horizon could not check.

dark matter haloes act to slow down bars via torques exerted through dynamical friction. bars slow down excessively. This has led to a growing tension...

Fragkoudi et al. 2021, 'Revisiting the tension between fast bars and the LCDM paradigm', as returned by this volume's search log, query 14 — the C15 occupier, verbatim

Four moves survived, and survived bounded — the bound is the claim, and it is stated as a claim: the searches did not find the discriminating statistic computed, and the horizon names the nearest neighbors it knows to be camped nearby. The response-lag signature survives with the territory active around it: the halo-response program (baryonic contraction and its successors, the Magellanic-Cloud wake papers), the

post-starburst kinematics program — eighty-nine post-starbursts already measured in the MaNGA survey — and the gas-sloshing literature, all occupying adjacent cells, none of them printing the three-way lag discriminator the signature requires. The residual census survives against a published null: the relation's residual scatter is on record as 0.057 dex, in agreement with modified dynamics, and no systematic correlation census of that scatter against bar strength, morphology, environment, gas fraction, and interaction stage surfaced in the searches or in the horizon's knowledge. The canalization import survives for want of any recovery-mode classification in the returns; and the analyticity check survives with the medium itself occupied — Blanchet's dipolar, gravitationally polarized dark matter is exactly the kind of medium the signature tests — but no dispersion-relation constraint on any published susceptibility in the returns. Four bounded survivors, every bound printed: this is what the source change bought, and the reader who produces any occupier converts a survivor into a kill by that act alone, the standing offer the series has always extended.

#	Query verbatim (abridged for the table; full text in the artifacts file)	State
1	Del Popolo 2024 time dependency of a_0 MOND acceleration scale	clean — the report's claim re-confirmed
2	galaxy dark matter halo response time delay after baryonic perturbation	clean — the response territory mapped
3	post starburst galaxy rotation curve dark matter response transient	degraded
4	radial acceleration relation residuals correlation morphology SPARC	degraded
5	bar pattern speed slowdown dark matter halo dynamical friction galaxies	junk
6	dark matter halo oscillation ringing after satellite impact recovery	junk
7	MOND gravitational polarization dielectric susceptibility response function	degraded
8	cosmic web filament bending stiffness elastic rods	junk
9	baryonic Tully-Fisher relation network optimization scaling derivation	degraded
10	Tisserand parameter dark matter capture galaxy halo phase space	degraded
11	persistent homology cosmic web topology dark matter discrimination	clean — C23's occupier found
12	consciousness adversarial collaboration IIT global workspace registered predictions results	degraded
13-2 2	the reworded second pass: residual scatter; bar slowdown; dipolar medium; Tully-Fisher feedback; Jacobi capture; COGITATE; filament dynamics; halo relaxation; halo sloshing; PSB kinematics	mixed — C15's occupier found (q14), the 0.057-dex null found (q13), 89-PSB program found (q22), sloshing mapped (q21); q17/q19 junk, q18 degraded

Table 7 — The occupation search log, twenty-two queries, seven degraded to junk or rate-limit noise and printed as degraded. The non-occupation claims of every surviving registration are bounded by this horizon, and the bound is part of the registration, not a footnote to it.

Survivor	Nearest neighbors, named	The unoccupied cell the signature holds
C13 / C16 (the response facet)	the halo-response program (baryonic contraction and successors; the Magellanic-Cloud wake); the 89-post-starburst MaNGA kinematics program; gas sloshing and cold fronts; the disk phase-spiral literature	the three-way lag-and-recovery discriminator: response immediate and monotone (collisionless), lagged and ringing (medium), or absent with the law algebraic (deletion) — on recently perturbed systems, timed against star-formation histories
C14 (the residual census)	the published null — 0.057 dex, in agreement with modified dynamics (Li 2018); the relation's original panels (Lelli-McGaugh-Schombert 2016)	the systematic correlation census: residual scatter against bar strength, morphology, environment, gas fraction, interaction stage — a five-probe matrix, pre-registered, on public tables, computable now
C20 (the analyticity check)	Blanchet's gravitational polarization program (the dipolar medium, 118 citations); the superfluid sector's phonon mediator; Verlinde's emergent family	the dispersion-relation constraint: whether any published susceptibility, extended to time-dependent response, satisfies causality's analyticity — a derivation check on archival equations, terminating at the literature itself

Table 8 — The survivors' nearest neighbors. The bound is printed with each survivor: these are the camps the horizon knows to be near, and the signature's claim is precisely the cell they have not computed.

*The residuals around these fits have an rms scatter of only 0.057 dex (13%).
This is in agreement with the predictions of modified Newtonian dynamics (MOND).*

Li et al. 2018, 'Fitting the radial acceleration relation to individual SPARC galaxies', as returned by this volume's search log, query 13 — the published null that registration R8 exists to test, verbatim

The convergence the pre-registration hoped for happened, and it happened in the survivors, not the dead: all four bounded survivors stand on one unstated assumption. C13 asks whether the response lags; C16 asks how it recovers; C20 asks whether the medium that produces it is causal in the dispersion-relation sense; C14 asks whether its residue is structured in the population. Four statistics, one facet — the dark sector's response function — and one fork printed in the next chapter with kill conditions on all sides. The favorite of the second generation is therefore the same shape as the favorite of the first: a three-way fork on an unstated core, ours harshest, dated. What differs is the unit: the first favorite forked on a number the knee tracks; the second forks on a response the field has never once asked to see.

5. The Registrations: The Response Fork and Its Companions

Three registrations are struck this volume, dated, with kill conditions on every side and the series' own side printed harshest, beside the six standing registrations carried forward from the first generation. The first and central one is the response fork — R7. It reads the dark sector's response function, the one object all four bounded survivors converge on, as a three-way discriminator with its own arithmetic: the dynamical time at the relation's knee is thirty to fifty million years for the galaxies the census actually holds, so a collisionless sector responds effectively instantaneously to any baryonic change slow enough to be called history; star-formation histories are timed to one hundred to five hundred million years; and the discriminating window is therefore the media whose relaxation times fall between a tenth of a gigayear and five — condensation-scale and sloshing-scale media, which is exactly the class the medium families propose. The fork's honest limit is printed with it: a fast medium is indistinguishable from no medium on lag alone, which is why the fork's second face — the recovery mode, ringing against monotone mixing — and its third face — the analyticity of the response — are carried in the same registration rather than as separate bets.

The response fork's sides	What resolves it	The kill condition, printed harshest for ours
(i) the response is immediate and monotone — the collisionless settlement's simplest face	recently perturbed systems' kinematics timed against their star-formation histories: interacting subsamples, post-starburst galaxies, the post-starburst programs already measuring 89 systems	dies to any lag or ringing in the 0.1-5 Gyr window above the timing floor; survives only if every response is complete within a dynamical time of its drive
(ii) the response lags and rings — a medium: condensation-scale or sloshing-scale relaxation	the same data, read as a distribution: response lag versus time since perturbation; recovery mode classified ringing versus monotone	dies to a census of post-perturbation systems showing responses complete within a dynamical time, with no phase structure — OURS dies with the first clean null
(iii) the response is absent — the law is algebraic, applied instantaneously with no medium at all (the deletion)	the residual census R8: if the deletion's residuals carry structure beyond the external field, the law's own scatter story fails; if they are pure noise, the law's prediction holds at the 2.2 percent floor	dies to structure: any of the five probes correlating above the floor without an external-field reading kills the simple-deletion story of its own residuals

Table 9 — R7, the response fork: the dark sector's response function as a three-way discriminator. Kill conditions on all three sides, ours (ii) harshest, resolvable on archival data now and on scheduled data through 2031; the conjunction with R1 rides with it.

The second registration is the residual census — R8 — and it is the series' first self-executable one: every input is public, every statistic is named, and the floor is computed and printed before anyone runs it. On a

sample of one hundred seventy-five galaxies with a residual scatter of 0.057 dex, the census detects any driver carrying at least 2.2 percent of the residual variance at ninety-five percent confidence; both programs predict structure above that floor — the settlement's feedback histories and formation channels, the deletion's external-field dependence — so a full null is the discriminating cell, not the expected one. The protocol is five probes against the residuals: bar strength, morphology, environment, gas fraction, and interaction stage, each pre-registered with its sign under each program. The published null it exists to test is on record saying the scatter is in agreement with modified dynamics; a census that finds the scatter structured by bars — or finds it structured by nothing at all — moves a live claim on public data, and the registration is dated the day before anyone looks.

R8's five probes	The settlement's expected sign	The deletion's expected sign
bar strength	correlated — bars exchange angular momentum with the live halo; the residual carries the exchange's history	null or weak — no dark sink; the external field dominates
morphology / surface brightness	correlated — formation channel and feedback history leave their mark in the residuals	weak — the law is claimed universal across morphology
cosmic-web environment	correlated — halo concentration and assembly bias vary with environment	correlated through the external field only — the direction differs from the settlement's
gas fraction	correlated — feedback couples gas to the halo's response	weak — gas enters only through the baryonic accounting
interaction stage	correlated — recent perturbations leave residual structure during relaxation	null after re-equilibration — the algebraic law has no relaxation channel

Table 10 — R8, the residual census: five probes, each pre-registered with its expected sign under each program, against the 2.2 percent-of-variance floor computed on the published inputs. The first registration in the series executable by anyone on public tables, without an instrument.

The third registration is the analyticity check — R9 — and it is the old-math lattice's bequest: a derivation-level test that terminates not on a telescope's schedule but on the literature's own published equations. Every medium family now standing in the dark sector's rival space — the dipolar, gravitationally polarized medium with its susceptibility; the superfluid sector with its phonon-mediated force; the emergent families with their effective responses — owes the record a response function, and a response function that describes a physical medium must be causal: no poles in the upper half plane, the real and imaginary parts locked together by the dispersion relations Kramers and Kronig wrote in 1926. The check is whether the published susceptibilities, extended to time-dependent response, satisfy that constraint — a question the searches found no one asking, on a medium the searches confirmed is one hundred and eighteen citations deep. A violation would mark an acausal medium; a pass would mark the first physical characterization of the sector's response in the frequency domain; either result is a line in the record, and the derivation is checkable by any disbelieving third party with the archival papers and

patience.

Zero survivors is a valid outcome, printed as the kill-log's result — the run's success metric is the registered, dated, checkable position on the record before the data are computed, NOT a surviving theory.

the pre-registration, worklog Task 15, locked 2026-09-30 and pushed at commit 7fac1ed before any search ran — the second generation's commitment, verbatim; four survivors are bounded, and the bound is the claim

Standing beside the three new registrations, the first generation's six carry forward unchanged, and one of them is now stronger than when it was struck: the conjunction cell that reads this volume's fork against the first fork — does the knee track $H(z)$ while the sector responds slowly, or does it track while the sector responds instantly — doubles the resolution of both, exactly as R4 doubled R1 and R3 by reading them together. And the bar dynamometer's corpse feeds the response fork directly: the field's own published tension — halos should brake bars hard, observed bars stay fast — is response-side evidence already on the record, the bequest rule applied, so R7 opens not as a speculation but as the first systematic statement of a question the field has already begun asking piecemeal without naming it.

6. The Next Sites: Consciousness Pre-Read, the Standing Order

The consciousness pre-read the report asked for is printed as a ruling the series can defend, not a prediction it can half make. The generator's site criterion has three clauses — a fresh census, an unspeakable core, and dated discriminating data — and the dark sector passes all three, which is why it was chosen and why it was held fixed for this volume's controlled test. Consciousness passes the first two: its assumption census would be rich (reports are transparent to experience; content and level are separable; the correlate is findable by contrast), and its unspeakable core needs no introduction. The third clause is where it fails as it stands: the field's discriminating instruments are adversarial collaborations and registered reports, and both the searches for their results and the horizon's own knowledge of them say the same thing — the confrontations are real, the biggest of them is already delivering results and already contested, and none of it is scheduled the way a survey is scheduled, with a date and a data-release architecture that a registration can be scored against. The searches meant to check this horizon degraded, both of them, and the pre-read prints that too: its bound is the bound of two lost queries and a horizon that knows the adversarial programs by name.

Site	The criterion as it stands	The verdict
the dark sector	fresh census; unspeakable core; dated discriminating data (DESI DR3, Euclid, Gaia DR4, the neutrino floor, CMB-S4, 2026-2031)	PASSED and held fixed for the controlled run — this volume
consciousness	the same three clauses; the dated-data clause asks for scheduled instruments that discriminate the live programs, and the field's best instruments are adversarial collaborations and registered reports on soft data	FAILS THE THIRD CLAUSE as it stands: the pre-read (this chapter's text) finds an adversarial-collaboration horizon (the integrated-information versus global-workspace confrontations, results arriving 2023-2026) that is real but weak — running generation would mean weakening the rule, and the weakening is a choice that must be printed before it is made, not after
quantum gravity	carried from Volume V: the strain is theory-internal; the counting rule's letter demands measured reproducible results, and the famous site fails first	CARRIED — until a measured thread exists

Table 11 — The next sites under the generator's own selection criterion. The order the record prints: the dark sector's controlled second generation (this volume); consciousness as an audit-first site with its rule change stated openly before any generation runs on it; quantum gravity carried.

The consequence is not a refusal but an ordering, and the ordering is already the report's own: audit first, then generate. A Volume-VIII audit of consciousness — observer mode, the verdict-emitter closed, the anomaly ledger built from the measurement programs the field actually runs — requires no weakening of anything, because the audit's discipline is to describe the strain, not to resolve it. Generation on consciousness is a different matter: it would run under a weakened dated-data clause, the weakening would have to be printed before the run rather than discovered after it, and this volume declines to make that choice silently in a chapter it has already used for other purposes. The record's own expected-value ledger adds the sober arithmetic the report's advice already guessed at: consciousness is the field with the most crowded idea-space the census has ever looked at — everyone can propose, no mathematics bars entry — and the thinnest signature-space, for the same reason; a generator that measures its yield in discriminating statistics should expect the dark sector's second-dozen pattern to look generous by comparison. The audit can still earn its keep: if the consciousness census finds load-bearing walls as stated as the dark sector's, the generator will know where the unstated floor is, and the out-of-frame lattices will know where to dig.

What this volume holds at its end, stated plainly, because the series grades itself before anyone else can: not a discovery, not a theory, and not even a claim that the four survivors are unoccupied — they are bounded, the bounds are printed with them, and every bound is an invitation to kill them. What is held is a method result and a position. The method result: the first generation's saturation was a property of the source, not of the domain — the census of a mature field is a map of occupied territory, the first generator sampled the map, and the second, sampling outside it, found the territory emptier than its map. The position: a three-way fork on the dark sector's response function, struck before anyone has computed

it, with the field's own fast-bar tension carried in as the first line of evidence; a residual census with its floor computed and its five probes pre-registered, executable by anyone on public tables; and an analyticity check on the medium families' own published equations. The verdict-emitter stays closed. The standing registrations from both generations resolve on the instruments' schedule through 2031, and the background process that scores them costs nothing and waits for no one. If a cell survives its kill condition, the series claims exactly the surviving cell — one line in the census's record, cited to these pre-registrations — and nothing more. The prose around a surviving cell may not be written until the cell resolves.

If you want real generation, change the input. Don't start from the current literature's frontier. Start from a deleted assumption, a cross-domain analogy, or a reinterpretation of old math. That's where P1, P2, and P6 from the original blueprint actually earn their keep.

the owner's situation report, this session (2026-09-30), verbatim — the redesign's charter, and the three lattices are its enumeration

12 moves pre-named through three out-of-frame lattices before any search ran — the pre-registration's lock	8 kills four by rule and arithmetic at generation, four by search with occupiers named, the favorite's neighbor first	1 facet four bounded survivors converging on the unstated assumption — the response fork, ours harshest, resolvable now
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This run's own numbers: twelve candidates pre-named through three out-of-frame lattices before any search ran; eight killed — four by rule and arithmetic at generation, four by search with occupiers named, the favorite's neighbor first; one facet surviving — four bounded signatures converging on the dark sector's response function, registered as a three-way fork with kill conditions on all sides, ours harshest. The bounds are the claim: produce an occupier and a survivor dies by that act alone. The verdict-emitter stays closed: registrations are not verdicts, and if a cell survives, the series claims exactly the cell — and nothing more. The ledger is open.